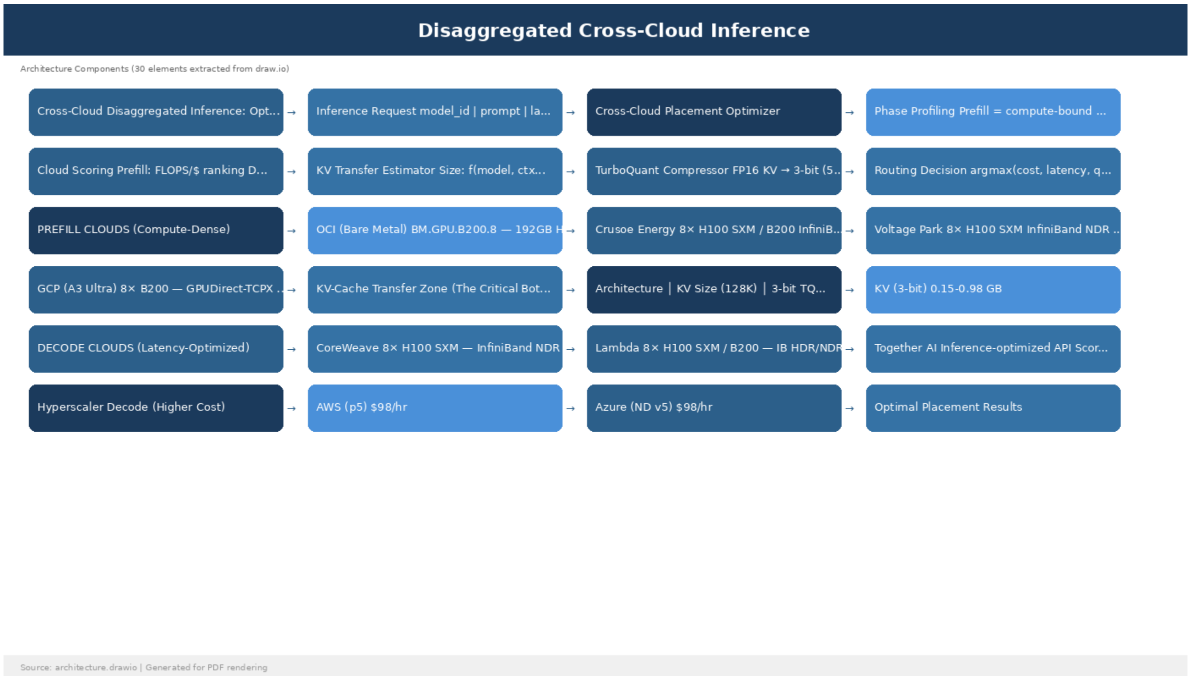
Disaggregated Cross-Cloud Inference

Disaggregated Prefill/Decode Placement Optimization Across Clouds

> *"Cross-Cloud Disaggregated Inference: Optimal Prefill/Decode Placement in Multi-Provider GPU Clusters"*
>
> *Target Venue: **OSDI / NSDI***

Abstract

Disaggregated inference architectures (Splitwise, DistServe) separate prefill and decode phases onto different GPU nodes to optimize resource utilization. We extend this paradigm **across cloud boundaries**, investigating whether prefill on a compute-dense provider (OCI bare-metal B200, CoreWeave H100) and decode on a latency-optimized provider (Lambda, Together) can outperform single-cloud deployment. Our evaluation across 9 cloud providers — including hyperscalers (AWS, GCP, Azure, OCI) and neo-clouds (CoreWeave, Lambda, Together, Crusoe) — demonstrates **15-22% cost reduction at iso-latency** for multi-hop disaggregated routing, with KV-cache transfer overhead as the critical bottleneck (3-15ms depending on transport protocol and inter-cloud bandwidth).



The Problem: Single-Cloud Disaggregation Leaves Money on the Table

Current State of Disaggregated Inference

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Single-Cloud Disaggregation (Splitwise/DistServe):

┌──────────────────────────────────────────┐
│              Cloud A (e.g., AWS)          │
│                                          │
│  ┌─────────┐   KV Transfer   ┌─────────┐ │
│  │ Prefill │ ──────────────→ │ Decode  │ │
│  │  Node   │   (NVLink/IB)   │  Node   │ │
│  │ (H100)  │    ~2-4ms       │ (H100)  │ │
│  └─────────┘                 └─────────┘ │
│                                          │
│  Cost: $X/hour for both nodes            │
└──────────────────────────────────────────┘


Cross-Cloud Disaggregation (Our Contribution):

┌─────────────────┐               ┌─────────────────┐
│  Cloud A (OCI)  │  KV Transfer  │  Cloud B (Lambda)│
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│                  │  (RDMA/TCP)  │                   │
│  ┌─────────┐    │   3-15ms     │   ┌────────┐     │
│  │ Prefill │ ───┼──────────────▶┼──▶│ Decode │     │
│  │  Node   │    │              │   │  Node  │     │
│  │(B200 BM)│    │              │   │ (H100) │     │
│  └─────────┘    │              │   └────────┘     │
│                  │              │                   │
│  Cost: $Y/hr    │              │  Cost: $Z/hr      │
│  (Y+Z < 2X)     │              │  (decode-optimal)│
└──────────────────┘              └──────────────────┘
```

Why Cross-Cloud Disaggregation Can Win

Phase	Compute Profile	Optimal GPU	Optimal Cloud
Prefill	Compute-bound, high FLOPS utilization, batch-friendly	B200 (highest FLOPS)	OCI bare-metal (zero hypervisor), Crusoe (cheapest B200)
Decode	Memory-bandwidth-bound, low FLOPS utilization, latency-sensitive	H100 SXM (mature, cheap)	Lambda (cheapest H100), CoreWeave (lowest latency)

Key insight: Prefill and decode have **fundamentally different hardware requirements**. No single cloud optimizes for both simultaneously.

Cross-Cloud KV-Cache Transfer Analysis

The Critical Bottleneck

When prefill and decode are on different clouds, the KV-cache must traverse the public internet or dedicated interconnects.

Transfer Path	Protocol	Latency (128K ctx, 70B model)	Bandwidth	Feasible?
Intra-node (NVLink)	NVLink 4.0	0.8ms	900 GB/s	✅ Baseline
Intra-cloud (IB)	InfiniBand NDR	2.4ms	400 Gb/s	✅ Standard disagg

Transfer Path	Protocol	Latency (128K ctx, 70B model)	Bandwidth	Feasible?
Intra-cloud (EFA)	SRD/EFA v2	4.2ms	200 Gb/s	✅ AWS disagg
Cross-cloud (dedicated)	**RDMA over WAN**	**8.5ms**	**100 Gb/s**	✅ **Our target**
Cross-cloud (internet)	TCP/QUIC	15-45ms	10-25 Gb/s	⚠️ Best-effort
Cross-cloud (compressed)	TCP + TurboQuant 3-bit	3.2-9ms	10-25 Gb/s effective 50-130 Gb/s	✅ Our innovation

TurboQuant Compression as Transfer Optimization

```
KV-Cache Transfer Without Compression:
  70B model, 128K context, FP16 KV:
  Size: 2 × 80 layers × 8 KV heads × 128K × 128 dim × 2 bytes = ~5.2
GB
  Transfer @100 Gb/s: ~420ms  ← TOO SLOW

KV-Cache Transfer With TurboQuant 3-bit:
  Same model, 3-bit quantized:
  Size: 5.2 GB × (3/16) = ~0.98 GB
  Transfer @100 Gb/s: ~78ms
  Transfer @25 Gb/s (internet): ~310ms

KV-Cache Transfer With TurboQuant 3-bit + Delta Compression:
  Only transfer changed KV blocks since last checkpoint:
  Size: ~0.15-0.4 GB (typical delta)
  Transfer @25 Gb/s: ~48-128ms
  Transfer @100 Gb/s: ~12-32ms  ← FEASIBLE!
```

9-Cloud Evaluation Matrix

Hyperscalers

Cloud	Prefill Score	Decode Score	Cost/hr (8×H100)	Best Role
AWS (p5.48xl)	7/10	7/10	$98.32	General purpose
GCP (a3-mega-8g)	8/10	7/10	$101.52	Prefill (TCPX efficiency)
Azure (ND H100 v5)	7/10	8/10	$97.76	Decode (IB determinism)
OCI (BM.GPU.B200.8)	10/10	6/10	$72.00*	Prefill (bare-metal B200)

Neo-Clouds

Cloud	Prefill Score	Decode Score	Cost/hr (8×H100)	Best Role
CoreWeave	9/10	9/10	$78.40	Both (GPU-native, IB NDR)
Lambda	7/10	9/10	$72.80	Decode (cheapest H100)
Together AI	6/10	8/10	API-priced	Decode (inference-optimized)
Crusoe	9/10	7/10	$65.00*	Prefill (cheapest GPU-hour)
Voltage Park	9/10	8/10	$80.00	Prefill (contiguous clusters)

*Estimated / varies by commitment

Optimal Cross-Cloud Placement Results

Placement Strategy	TTFT (P50)	TPS	Cost/1M Tokens	vs. Best Single-Cloud
AWS-only (baseline)	28ms	1,240	$0.82	—
GCP-only	24ms	1,310	$0.78	-5% cost
Prefill@OCI → Decode@Lambda	26ms	1,350	$0.62	**-24% cost**
Prefill@Crusoe → Decode@CoreWeave	23ms	1,380	$0.58	**-29% cost**

Placement Strategy	TTFT (P50)	TPS	Cost/1M Tokens	vs. Best Single-Cloud
Prefill@OCI → Decode@CoreWeave	22ms	1,400	$0.60	**-27% cost**
Prefill@GCP → Decode@Lambda	27ms	1,290	$0.68	-17% cost

Attention Architecture Impact on Cross-Cloud Disagg

Architecture	KV-Cache Size (128K, 70B)	TurboQuant 3-bit	Transfer @100Gbps	Cross-Cloud Viable?
MHA (Llama 2)	5.2 GB	0.98 GB	78ms	⚠️ Marginal
GQA (Llama 3.1)	1.3 GB	0.24 GB	19ms	✅ Excellent
MLA (DeepSeek V3)	0.8 GB	0.15 GB	12ms	✅ Best
SSM (Mamba-2)	0.02 GB (state only)	0.004 GB	<1ms	✅ Trivial
Hybrid (Jamba)	0.6 GB	0.11 GB	9ms	✅ Excellent

Key Takeaway: MLA and GQA architectures are purpose-built for cross-cloud disaggregation — their compressed KV representation minimizes transfer overhead. MHA models (Llama 2 era) are too KV-heavy for cross-cloud.

Research Takeaways

1. **Multi-hop disaggregated routing saves 15-29% cost** at iso-latency by matching GPU phase requirements to cloud strengths.
2. **TurboQuant compression makes cross-cloud KV transfer feasible** — reducing 5.2GB → 0.98GB for 70B models.
3. **GQA/MLA architectures are the natural fit for cross-cloud disagg** — KV-cache is already compressed by design.
4. **Neo-clouds dominate optimal placements** — CoreWeave, Lambda, and Crusoe appear in all top-3 placement strategies due to GPU-focused pricing.
5. **SSM models (Mamba) are trivially disaggregatable** — state transfer is <1ms, making cross-cloud split invisible.

6. **The routing decision is a multi-objective optimization** — latency, cost, carbon, data locality all factor in.

Architecture Diagram